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Dr. Seussifying research on real-time dream communication

The Research Article

Reports of dreams from individuals after they have woken up can be difficult to use for research as they can often be distorted due to the individual forgetting or misremembering their dream. This poses a challenge for dream research, as accurate data can be hard to acquire. However, with a team of 20 researchers from the United States, France, Germany and the Netherlands, Konkoly et al. (2021) provided evidence for a method to facilitate two-way communication with a lucid dreaming individual in real-time.

Two important prerequisites for the communication to take place were that the sleeping individual had to be in REM sleep and that they had to be lucid dreaming. Consequently, only individuals with a history of being able to lucid dream were involved with the study. While determining whether the individual had entered REM sleep was done with measuring EEG activity[1], the individuals themselves had to let the researchers know when they entered a lucid dreaming state using a series of left-right eye movements, which were picked up as EOG signals. Once both of these conditions were met, the researchers would ask simple yes-no or arithmetic questions by using either their voice or by shining light and playing tones in morse code. These were then answered by the test participants using either eye movements picked up as EOG signals or muscle contractions picked up as EMG activity. The teams in the US, Germany and the Netherlands all used eye movements as output signals, while only the French team used facial muscle contractions.

A total of 36 individuals participated in the experiment which included 57 sessions of REM sleep. During these sessions, an attempt at communication was made 158 times with varying success. While there were 29 correct responses (18.4%), there were also 5 incorrect (3.2%) and 28 ambiguous (17.7%) responses. During the remaining 96 times (60.8%), no response was measured. Still, Konkoly et al. have provided solid evidence to suggest that two-way communication in real-time with dreaming individuals is a replicable phenomenon, which can become a new strategy for the research of dreams in the future.

[1] To be as sure as possible whether or not REM sleep had begun, external doctors were recruited to make independent judgments. Only sessions where a majority of these doctors stated that they qualified as REM sleep were focused on to determine the legitimacy of two-way communication.

The Chosen Approach

During my personal studies, I often struggle with processing scientific articles due to the complexity of the language and terms used. This leads to me having to reread sections multiple times, ultimately costing a lot of time and energy. The same can be said about the research article chosen for this project. Inspired by this experience, I set out to express the outcomes of this research as simple as possible. The manner in which I have decided to do so was to take the gist of the outcome of the article and tell it in the form of a children's book in the style of Dr. Seuss. This meant simplifying the subject matter to the point a child could understand it.

Translating a research article into a Dr. Seuss-esque children's book comes with a number of challenges, the first of which being the writing. When trying to replicate Seuss' writing style, one needs to keep in mind the characteristics that make a Dr. Seuss rhyme sound like a Dr. Seuss rhyme. A big part of this comes in the form of simplicity and repetition. Suess often repeats large parts of certain sentences, only changing one or two words. Sometimes, full sentences are repeated without any modification at all. A great example can be found in "*Green Eggs and Ham*" (1960), which opens with the following repetitive rhyme:

> *"That Sam-I-Am!*
> *That Sam-I-Am!*
> *I do not like*
> *That Sam-I-am!"*

"*Green Eggs and Ham"* is also representative of another one of Seuss' writing style quirks. For some of his books targeted at younger children, Seuss used almost no words that were longer than one syllable. Including *"Green Eggs and Ham"*, which only includes one word longer than one syllable, other examples of this focus on monosyllabic writing are "*The Cat in the Hat" (1957)* and "*One Fish Two Fish Red Fish Blue Fish*" (1960).

While there are other traits of Dr. Seuss' writing that are emblematic to him, I chose to focus on keeping my rhymes simple, using repetition frequently and using mostly monosyllabic words, which was restrictive enough in its own right.

Another aspect of Dr. Seuss' books are his whimsical illustrations. His usage of black outlines and solid, bright colours in combination with his way of drawing furry characters have become synonymous with the Dr. Seuss art style. Furthermore, Dr. Seuss' backgrounds often depict surreal architecture and fantastical nature landscapes.

Taking these aspects of Dr Seuss' works into consideration, I wrote my own story which tries to communicate the outcomes of the chosen research article.

Wig Wush and Pat: The Tale of the Dream Hat

The book's story is about a small bug named Wig Wush and his best friend Pat, a brown bear. Wig Wush and Pat love to talk to each other and do so constantly. However, as Pat wants to go to sleep, which would end their talk, Wig Wush presents Pat with the titular *Dream Hat*. The *Dream Hat* allows Pat to communicate with Wig Wush while he is dreaming. While wearing the hat, Pat can flex either the muscle in his cheek or eyebrow to indicate a yes or no respectively. This result is then relayed through the hat to Wig Wush, by shining either a green or red light. This is analogous to the research article's method of allowing dreamers to respond to the experimenter's questions by contracting either their zygomatic or corrugator muscle, which are located near the cheek and eyebrow. The *Dream Hat* itself is an analogy for the EMG electrodes connected to these muscles. By showcasing the ability to have two-way communication with a dreamer in real-time, this work attempts to convey the main message of the chosen research article.

Wig Wush and Pat: The Tale of the Dream Hat has 228 unique words, of which only seven (3%) are not monosyllabic. Furthermore, as mentioned above, repetition was utilized in various places to get closer to Dr. Seuss' writing. In certain cases, artificial intelligence chatbots were used to help get suggestions for rhymes.

Due to my lacking artistic ability, I made to choice to limit the number of illustrations. Consequently, I also chose to format the work as a "webtoon", meaning that the pages are thin and long. This helped minimize empty space. The illustrations that are included are either traced images from existing works of Dr. Seuss with slight modifications, or simply copy pasted from those works. While using generated AI art was considered, the idea was quickly discarded due to the unsatisfactory quality of the illustrations.

Bibliography:

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